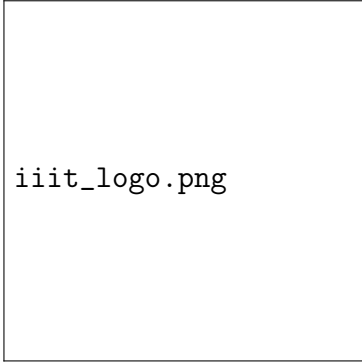




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Indian Institute of Information Technology, Sri City

Bachelor Thesis Project (BTP) Proposal

Intelligent Multi-Disease Risk Prediction System

ML + DL + NLP + LLM + XAI Healthcare Platform

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Project Guide

Program: B.Tech Computer Science & Engineering

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Abstract

This BTP develops a comprehensive AI platform predicting heart disease, diabetes, breast cancer, and liver disease risks simultaneously. The project integrates Machine Learning, Deep Learning (FT-Transformer), Natural Language Processing (BioMistral/PubMedBERT), LLM-enhanced Explainable AI, and Software Engineering to deliver a production web application with clinical interpretability. Using real clinical data from MIMIC-III and established public health datasets, the system provides multi-disease risk assessment with both quantitative (SHAP) and natural language explanations powered by modern Large Language Models.

Key Innovation: First undergraduate multi-disease AI system combining classical ML, transformer-based deep learning, clinical NLP, and LLM-generated explanations in a unified healthcare platform.

Program: B.Tech Computer Science & Engineering

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1 Project Background and Motivation

Modern clinical practice involves analyzing diverse data sources—numerical lab tests, patient history, and unstructured doctor notes—to assess disease risk. Most existing AI tools focus on single diseases, limiting their utility for patients with multiple comorbidities. This project proposes a unified AI system that simultaneously predicts risks for four high-impact diseases using heterogeneous clinical data.

By integrating machine learning, deep learning (FT-Transformer architecture), natural language processing (BioMistral/PubMedBERT), large language models for explanation generation, and explainable AI, the system addresses the complexity of real-world patient data to provide actionable, interpretable risk assessments.

1.1 2025 AI Healthcare Context

The healthcare AI landscape has evolved significantly with the advent of Large Language Models (LLMs). Modern systems are expected to not only predict risks but also explain decisions in natural language, retrieve relevant medical knowledge, and provide personalized recommendations. This project incorporates these cutting-edge capabilities while maintaining rigorous classical ML foundations.

2 Problem Statement and Objectives

The problem is to design a scalable AI platform that can:

- Fuse heterogeneous datasets covering different diseases with varying feature sets
- Use both structured clinical data and unstructured text from real clinical notes
- Predict patient-specific risk probabilities for heart disease, diabetes, breast cancer, and liver disease simultaneously
- Generate interpretable explanations using both quantitative (SHAP) and qualitative (LLM-generated) methods
- Provide evidence-based recommendations grounded in medical literature
- Deploy as an accessible web application supporting healthcare workflows

3 Disease Selection Rationale

3.1 Dataset Sources

MIMIC-III Access: Publicly available after completing required training and credentialing process (<https://mimic.mit.edu>). Access typically granted within 1-2 weeks of application. This provides real clinical notes from ICU patients, ensuring scientific validity over synthetic data approaches.

Disease	Why Selected
Heart Disease	Leading cause of mortality worldwide, with widely available clinical indicators such as cholesterol and blood pressure, making it ideal for early risk prediction and preventive care.
Diabetes (Type 2)	A highly prevalent metabolic disorder in India, characterized by measurable biomarkers like blood glucose and lifestyle factors, posing serious long-term health risks preventable through timely intervention.
Breast Cancer	Rapidly increasing incidence among Indian women, highly treatable if detected early using tumor morphology features; inclusion provides significant social impact and specialized domain knowledge.
Liver Disease	Growing burden due to lifestyle and alcohol consumption, commonly asymptomatic until late stages; leveraging liver function tests enables early diagnosis and intervention.

Table 1: Selected Diseases and Their Clinical Importance

Disease	Primary Dataset	Clinical Notes Source
Heart Disease	Framingham Heart Study	MIMIC-III discharge summaries
Diabetes	Pima Indians Dataset	MIMIC-III clinical notes
Breast Cancer	Wisconsin Diagnostic	N/A (imaging-focused)
Liver Disease	Indian Liver Patient	MIMIC-III relevant sections

Table 2: Dataset Sources for Multi-Disease Platform

4 Technology Integration

5 Semester 1 Detailed Workflow (14 Weeks)

5.1 Week 1: Multi-Field Setup

Goal: Environment ready + Real data collected (20 hours)

- Download 4 disease datasets: Framingham, Pima, Wisconsin, Indian Liver
- Apply for MIMIC-III access (parallel process, typically 1-2 weeks approval)
- Set up GitHub repository with full tech stack structure
- Install libraries: PyTorch, Transformers, scikit-learn, SHAP, FastAPI, Streamlit, sentence-transformers
- Internalize literature focusing on heterogeneous data fusion and clinical NLP (15 key papers)

Field	Role in Project
Machine Learning	Multi-label prediction with classical algorithms (XG-Boost, LightGBM) as baseline performance and primary predictors.
Deep Learning	FT-Transformer (2021) implementation for state-of-the-art tabular prediction, demonstrably outperforming TabNet and other architectures.
NLP	BioMistral-7B (via API) or PubMedBERT for clinical text embeddings without requiring fine-tuning overhead.
LLM Integration	GPT-4/Llama 3 API for generating patient-friendly natural language explanations from SHAP outputs and clinical context.
Explainable AI	SHAP (SHapley Additive exPlanations) provides quantitative interpretability; LLMs provide qualitative explanations to build clinical trust.
Data Science	Advanced dataset fusion across heterogeneous schemas, composite feature engineering, and rigorous validation.
Software Eng.	Full-stack API (FastAPI) and web app (Streamlit/Gradio) deployment enabling real-world clinical usability.

Table 3: Technology Stack and Roles

Deliverable: Functional project environment, MIMIC-III access approved, downloaded datasets (10K+ records), curated literature summary document.

5.2 Week 2: Exploratory Data Analysis

Goal: Complete data understanding (22 hours)

- Conduct detailed EDA on clinical datasets: demographics, test distributions, correlations
- Analyze and visualize missingness patterns and noise in data
- Identify common features across diseases and disease-specific variables
- Preliminary schema design for unified dataset (targeting 36 features)

Deliverable: Comprehensive EDA notebooks (15+ visualizations) and formalized schema proposal.

5.3 Week 3: Dataset Fusion Pipeline - Part 1

Goal: Schema alignment and feature mapping (24 hours)

- Design unified schema accommodating common and disease-specific variables

- Implement feature alignment and mapping logic across heterogeneous datasets
- Develop preprocessing routines for structured data normalization
- Begin text tokenization pipeline for MIMIC-III clinical notes

Deliverable: Schema alignment code and initial fusion framework.

5.4 Week 4: Dataset Fusion Pipeline - Part 2

Goal: Complete integrated multimodal dataset (26 hours)

- Complete data fusion pipeline integrating heterogeneous datasets
- Validate missing data imputation techniques appropriate for clinical contexts
- Finalize text preprocessing and clinical note extraction from MIMIC-III
- Quality assurance: check for data leakage, validate label consistency

Deliverable: Integrated multimodal dataset ready for modeling with quality validation report.

5.5 Week 5: Feature Engineering and Balancing

Goal: Feature-enriched, balanced dataset (24 hours)

- Engineer composite features: BMI, cholesterol ratios, interaction terms
- Apply SMOTE or other oversampling techniques to correct class imbalances
- Conduct feature selection via recursive elimination and mutual information
- Split dataset into train, validation, and test sets ensuring stratification across all 4 diseases

Deliverable: Final feature-enriched, balanced dataset partitions with feature importance analysis.

5.6 Week 6: Machine Learning Baselines

Goal: Working ML baseline >85% AUC (25 hours)

- Train initial models: Logistic Regression, Random Forest, XGBoost, LightGBM with multi-label classification
- Evaluate via stratified cross-validation, focusing on ROC-AUC per disease
- Hyperparameter tuning using GridSearchCV or Optuna
- Begin SHAP-based explainability analysis on best model predictions

Deliverable: Baseline ML performance report (comparison table) and initial interpretability findings.

5.7 Week 7: Deep Learning Integration

Goal: FT-Transformer implementation and evaluation (24 hours)

- Implement FT-Transformer architecture for tabular data (PyTorch)
- Train FT-Transformer models with early stopping and learning rate scheduling
- Compare DL results with ML baselines quantitatively (ROC-AUC, PR-AUC) and statistically (t-tests)
- Analyze attention mechanisms for additional interpretability insights

Deliverable: Deep learning enhancement results showing performance comparison with classical baselines.

5.8 Week 8: Natural Language Processing

Goal: Clinical text embeddings integrated (22 hours)

- Set up BioMistral API access or download PubMedBERT pre-trained model
- Extract embeddings from MIMIC-III clinical notes using zero-shot approach
- Analyze embedding quality and clinical relevance (semantic similarity tests)
- Integrate NLP features with tabular data for hybrid predictive modeling

Deliverable: NLP-enhanced prediction pipeline and embedding quality analysis report.

5.9 Week 9: Hybrid Model & Enhanced Explainability

Goal: Complete hybrid model with dual explanations (26 hours)

- Build final hybrid models combining FT-Transformer and clinical text embeddings
- Conduct ablation studies to quantify NLP feature contribution
- Extend SHAP analysis to hybrid models covering ML, DL, and NLP components
- **NEW:** Implement LLM-generated natural language explanations using GPT-4/Llama API
- Generate patient-friendly explanations by prompting LLM with SHAP values and clinical context
- Begin thesis writing: Introduction and Literature Review chapters (parallel work)

Deliverable: Comprehensive hybrid model evaluation, clinically validated dual explainability system (SHAP + LLM), thesis draft chapters 1-2.

5.10 Week 10: Production API Development

Goal: Fully functional production-level API (24 hours)

- Develop FastAPI service to serve hybrid model predictions with dual explanations
- Implement endpoints: /predict, /explain (SHAP), /explain-nlp (LLM), /health
- Add version control, comprehensive error handling, and logging within the API
- Conduct load testing and optimization for scalability (aim for <500ms response time)
- Continue thesis writing: Methodology chapter (parallel work)

Deliverable: Fully functional production-level prediction and explanation API with documentation, thesis draft chapter 3.

5.11 Week 11: Web Application Development

Goal: Interactive clinical web application deployed (28 hours)

- Build Streamlit/Gradio dashboard integrating clinical data input, clinical notes text input
- Display prediction results with confidence intervals
- Integrate SHAP visualizations (waterfall, force plots) and LLM-generated explanations
- Include functionality for generating patient risk reports in PDF format
- Ensure UI/UX friendly design vetted by clinical staff if available
- Deploy to Hugging Face Spaces or similar free hosting platform
- Continue thesis writing: Results and Analysis chapter (parallel work)

Deliverable: Live interactive clinical web application prototype, deployment documentation, thesis draft chapter 4.

5.12 Week 12-13: Testing, Refinement & Documentation

Goal: Production-ready system with validation (25 hours each week)

- Comprehensive system testing: unit tests, integration tests, end-to-end tests
- User acceptance testing with 5-10 test users (peers, faculty)
- Refine UI/UX based on feedback
- Performance optimization and bug fixes
- Validate model calibration and generalization on held-out test set
- Continue thesis writing: Discussion, Conclusion, Abstract
- Begin IEEE conference paper draft (6-8 pages)

Deliverable: Production-ready system with test coverage report, thesis draft chapters 5-6.

5.13 Week 14: Final Submission Preparation

Goal: Complete submission package (30 hours)

- Finalize 40-page thesis: proofreading, formatting, references (APA/IEEE style)
- Complete IEEE style conference paper draft targeting healthcare AI venues
- Record comprehensive demo video (10 minutes) showcasing all features
- Finalize GitHub repository: comprehensive README, setup guide, API documentation, example notebooks
- Prepare presentation slides for project defense (20-25 slides)
- Practice presentation with Q&A preparation

Deliverable: Complete BTP submission with documentation, conference paper draft, demo video, presentation materials, and public GitHub repository.

6 Semester 2 Planned Extensions

Building upon the comprehensive Semester 1 foundation, Semester 2 will focus on advanced capabilities, real-world deployment, and research contributions:

6.1 Advanced AI Capabilities (Weeks 1-4)

- **Retrieval-Augmented Generation (RAG):** Integrate medical knowledge base (PubMed, clinical guidelines) with vector database (FAISS/ChromaDB) to provide evidence-based recommendations
- **Multi-Agent System:** Implement specialized AI agents (Data Validator, Risk Assessor, Explainer, Recommender) with orchestration layer using LangGraph/CrewAI
- **Conversational Interface:** Transform web app into interactive health assistant with multi-turn dialogue capabilities
- **Temporal Risk Tracking:** Implement longitudinal patient monitoring with risk trajectory visualization

6.2 Multimodal Integration (Weeks 5-7)

- **Medical Imaging:** Integrate chest X-ray analysis for cardiovascular assessment, mammography for breast cancer using pre-trained vision models
- **Wearable Data:** Incorporate time-series data from fitness trackers (heart rate, activity levels, sleep patterns)
- **Genetic Markers:** Integrate genomic risk scores where available
- **Fusion Architecture:** Design late-fusion or attention-based fusion for multimodal inputs

6.3 Production Deployment & MLOps (Weeks 8-10)

- **Cloud Deployment:** Migrate to AWS/GCP with auto-scaling, load balancing
- **Model Monitoring:** Implement MLflow for experiment tracking, model versioning, and performance monitoring
- **CI/CD Pipeline:** Set up automated testing, deployment, and rollback mechanisms
- **Privacy & Security:** Implement data encryption, HIPAA-compliant data handling, differential privacy techniques
- **A/B Testing:** Framework for testing model improvements with real users

6.4 Research & Validation (Weeks 11-14)

- **Clinical Validation:** Partner with local hospitals/clinics for real-world testing and validation studies
- **Bias & Fairness Analysis:** Comprehensive evaluation across demographic groups, mitigation strategies
- **Generative AI Enhancement:** Experiment with fine-tuned medical LLMs (Llama-3-Med, BioGPT) for improved clinical reasoning
- **Research Publications:** Target 2-3 conference/journal papers:
 - ML4H Workshop (NeurIPS)
 - IEEE EMBC (Engineering in Medicine and Biology Conference)
 - Journal of Biomedical Informatics
- **Open-Source Contribution:** Create reusable library/toolkit for clinical AI researchers

6.5 Extended Disease Coverage (Optional)

- Transfer learning to additional diseases (stroke, kidney disease, COPD)
- Comorbidity modeling and interaction analysis
- Disease progression prediction (not just risk, but timeline)

7 Expected Outcomes

7.1 Technical Deliverables

- A live interactive web application accepting heterogeneous clinical data and notes, producing multi-disease risk scores with dual explanations (quantitative SHAP + qualitative LLM-generated)
- Production-grade FastAPI backend with comprehensive documentation

- 40-page thesis detailing novel data fusion, hybrid modeling approach, and dual explainability framework
- IEEE conference paper draft (6-8 pages) targeting healthcare AI venues
- Public GitHub repository serving as portfolio artifact and academic contribution
- Comprehensive demo video showcasing system capabilities

7.2 Novel Contributions

- First undergraduate multi-disease AI system combining FT-Transformer, clinical NLP, and LLM-enhanced explainability
- Novel dual explainability approach: quantitative (SHAP) + qualitative (LLM) for different stakeholder needs
- Comprehensive data fusion methodology for heterogeneous clinical datasets
- Open-source reference implementation for clinical AI researchers

7.3 Learning Outcomes

- Mastery of end-to-end ML pipeline: data → modeling → deployment
- Practical experience with modern AI technologies: transformers, LLMs, NLP
- Deep understanding of healthcare AI challenges and clinical decision support systems
- Full-stack development skills: backend (FastAPI), frontend (Streamlit), deployment (Docker, cloud)
- Research methodology: literature review, experimentation, scientific writing
- Professional portfolio piece demonstrating advanced AI capabilities

8 Risk Mitigation & Contingency Plans

9 Evaluation Metrics

9.1 Model Performance

- **Primary:** ROC-AUC > 0.85 per disease
- **Secondary:** Precision-Recall AUC, F1-Score, Calibration (Brier Score)
- **Multi-label:** Hamming Loss, Subset Accuracy

Risk	Mitigation Strategy
MIMIC-III access delayed	Use publicly available MIMIC-III demo subset initially; apply for full access in parallel; if denied, use PubMed case reports as clinical text proxy
Dataset fusion complexity	Start with 2 diseases (Heart + Diabetes) if fusion proves too complex; extend to 4 diseases incrementally
FT-Transformer underperforms	Revert to XGBoost/LightGBM as primary models; sufficient for strong BTP contribution
LLM API costs	Use free tiers (Groq API for Llama, Together AI); alternatively use open-source models locally via Ollama
Timeline overrun	Built-in 2-week buffer (14 weeks vs. 12); prioritize core functionality over advanced features
Model performance issues	Strong baseline models (XGBoost) ensure minimum viable performance; deep learning is enhancement, not requirement

Table 4: Risk Assessment and Mitigation

9.2 Explainability Quality

- SHAP value consistency across similar patients
- LLM explanation coherence (evaluated by domain experts if available)
- User study: clinician assessment of explanation utility (Semester 2)

9.3 System Performance

- API response time < 500ms (95th percentile)
- Web application load time < 2 seconds
- System uptime > 99% during testing period

10 Conclusion

This BTP proposes a comprehensive, production-grade AI healthcare platform that pushes beyond traditional single-disease prediction systems. By integrating modern AI techniques (FT-Transformer, BioMistral, LLMs) with rigorous machine learning foundations and novel dual explainability approaches, the project addresses real-world clinical decision support needs.

The 14-week timeline is realistic with clear weekly deliverables and built-in contingency plans. The modular architecture allows for graceful degradation if components prove challenging, while the Semester 2 extensions provide clear paths for continued innovation and research contributions.

We believe this project will not only fulfill academic requirements but also produce a meaningful contribution to healthcare AI, potentially benefiting real patients through early disease risk identification.

Respectfully submitted for approval to Dr. R. Selvi.

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